SYNAPSE EDGE DEFENSE

Deployment Topology

One control plane. A fleet of edge sensors. A persistent WebSocket tunnel that carries config in and telemetry out — with tenant isolation enforced at every layer.

1	N	5	Multi
CONTROL PLANE	EDGE POPS	TUNNEL CHANNELS	TENANT ISOLATED

PHYSICAL & LOGICAL TOPOLOGY

CONTROL-PLANE SERVICES

RuleDistributor
FLEET RULE SYNC

Tracks per-sensor rule status, enforces tenant isolation, coordinates blue/green swaps on rule updates.

RolloutOrchestrator
BATCHED ROLLOUTS

Health-aware rollout of binary releases via BullMQ queue. Fails a batch early rather than rolling the whole fleet.

FleetCommander
REMOTE CONTROL

Routes reload/restart/drain/resume commands to named sensors over the tunnel. Audit-logged.

TunnelBroker
CHANNEL MULTIPLEX

WebSocket session manager. Per-session auth, per-channel rate limits, per-sensor concurrent session cap.

Telemetry API
EVENT INGEST

JWT-auth'd batched ingest with replay protection. Feeds ClickHouse & materialized views.

DeploymentStateStore
BLUE/GREEN STATE

Durable record of which rule set each sensor is serving — survives control-plane restarts.

RULE & RELEASE PROPAGATION — HEALTH-AWARE BATCHING

1%
CANARY
One sensor per PoP gets the new rule set first. Block on health check pass.

10%
EARLY WAVE
Expands to a single region after canary green. First look at real traffic patterns.

50%
HALF FLEET
Blue/green split. Failed batches stop the rollout; passing batches hold.

100%
FULL ROLLOUT
Remaining sensors swap. Previous state retained for instant rollback via the state store.

TUNNEL PROTOCOL — 5 MULTIPLEXED CHANNELS

Shell	Logs	Diag	Control	Files
Remote terminal with PTY	Live log streaming w/ filters	Health · mem · rules · actors	reload · restart · drain	Secure file xfer & browse

Tenant isolation at every layer

Sensors belong to exactly one tenant. Rule distribution, tunnel sessions, and telemetry queries all enforce tenant scoping — attempting to touch another tenant's sensors throws `TenantIsolationError` before any side effect occurs. Multi-tenant deployments share the control plane, not the data.

One tunnel in. Tenant-isolated all the way down.

Health-aware rollouts. Blue/green rule swaps. Instant rollback. Every channel audit-logged.